National Academic Test Program
โครงการสอบวัดทักษะวิชาการระดับชาติ (สวช.)

การแข่งขันสอบวัดทักษะวิชาการระดับชาติ ประจำปี พ.ศ. 2563

วิชาภาษาอังกฤษ ระดับชั้นมัธยมศึกษาตอนปลาย

รหัสวิชา อ405 ภาษาอังกฤษ

รหัสชุดข้อสอบ 405002

ชื่อ.....นามสกุล.....รหัสวิชา.....

ระดับชั้น.....ห้องสอบ.....วันที่สอบ.....สถานที่สอบ.....

คำชี้แจง

รายละเอียดข้อสอบ	**ข้อสอบฉบับนี้มี 11 หน้า จำนวน 50 ข้อ ระยะเวลา 90 นาที**		
วิธีการตอบ	**ให้ใช้ดินสอ 2B ระบายในวงกลมที่เป็นคำตอบในกระดาษคำตอบ**		
เกณฑ์การให้คะแนน	**ส่วน Expression**	**ข้อ 1-10**	**ข้อละ 1.6 คะแนน**
(คะแนนเต็ม 100 คะแนน)	**ส่วน Reading Comprehension**	**ข้อ 11-25**	**ข้อละ 2.4 คะแนน**
	ส่วน Structure	**ข้อ 26-40**	**ข้อละ 2.2 คะแนน**
	ส่วน Vocabulary	**ข้อ 41-50**	**ข้อละ 1.5 คะแนน**

ข้อปฏิบัติในการสอบ

1. ให้ผู้เข้าสอบปฏิบัติตามระเบียบสำนักงานโครงการสอบวัดทักษะวิชาการระดับชาติ (สวช.) ภาคีเครือข่ายโรงเรียนทั่วประเทศ และสถาบันอีซี่ลิชประเทศไทย ว่าด้วยแนวทางปฏิบัติเกี่ยวกับการดำเนินการสอบ พ.ศ. 2560 อย่างเคร่งครัด
2. กรอกข้อมูลส่วนตัวและข้อมูลเกี่ยวกับการสอบให้ครบถ้วนบนหน้าปกแบบข้อสอบและในกระดาษคำตอบ
3. อ่านวิธีการตอบข้อสอบให้เข้าใจแล้วตอบข้อสอบด้วยตนเอง และไม่เอื้อให้ผู้อื่นคัดลอกคำตอบได้
4. ห้ามคัดลอก บันทึกภาพ หรือเผยแพร่ข้อสอบก่อนมีการเฉลยโดยเด็ดขาด
5. ไม่อนุญาตให้ผู้เข้าสอบออกจากห้องสอบก่อน 1 ชั่วโมงหลังจากเริ่มเวลา
6. ไม่อนุญาตให้ผู้คุมสอบเปิดอ่านข้อสอบ

หากผู้เข้าสอบฝ่าฝืนข้อปฏิบัติ สำนักงาน สวช. จะมีการดำเนินการ ดังนี้

1. ไม่ประกาศผลการสอบจากการทดสอบดังกล่าว
2. แจ้งไปยังโรงเรียนต้นสังกัดของนักเรียน ผู้ดูแลของผู้สอบเพื่อดำเนินการตักเตือนและกล่าวโทษ
3. แจ้งไปยังผู้บริหารสถานศึกษาที่เป็นภาคีเครือข่าย สวช. เพื่อพิจารณารับเข้าศึกษา

เอกสารนี้เป็นลิขสิทธิ์ของสำนักงานโครงการสอบวัดทักษะวิชาการระดับชาติ (สวช.) และสถาบันอีซี่ลิชประเทศไทย บริษัท อีซี่ลิช อินเตอร์-เอ็ดดูเคชั่น (ประเทศไทย) จำกัด

การทำซ้ำหรือดัดแปลงหรือเผยแพร่เอกสารฉบับนี้จะถูกดำเนินคดีตามกฎหมาย

Part I: Expression (Items 1 – 10)
Choose the best answer.

Situation: On the street (Items 1 – 5)

Pete: Good afternoon, Chris. It's been a while. How have you been?
Chris: Good afternoon, Pete. _____(1)_____. What about you?
Pete: Couldn't be better as well. What are you up to?
Chris: I am going to the hospital.
Pete: _____(2)_____.
Chris: I'm going to pick up my wife there.
Pete: What's the matter with her?
Chris: She has trouble sleeping at night. She has always got a headache and never feels fresh.
Pete: Does she drink coffee?
Chris: I don't think so. _____(3)_____. Anyway, she seems extremely stressed about her work these days.
Pete: That's why she has trouble fall asleep.
Chris: What do you mean?
Pete: I mean, a person who get stressed all the day can cause insomnia.
Chris: Oh dear, I hope she will not be.
Pete: _____(4)_____. Anyway, why don't you ask her to exercise regularly either in the morning or in the evening?
Chris: You may be right. I have also heard that it is a helpful treatment.
Pete: _____(5)_____**,** she will get well without a doctor in no time.
Chris: From tomorrow morning, I'll tell her to start taking exercise.
Pete: That is a good thing. Good luck with that. Good bye.
Chris: Thank you so much. Good bye.

1. 1. I couldn't agree at all. 2. I have been doing very well. 3. I am doing fine. 4. I'd better not. 5. I'm so glad to meet you.

2. 1. In what connection? 2. What's going on? 3. How can it be? 4. What hospital are you going? 5. Can I come with you?

3. 1. I see her drinking coffee every day. 2. She loves having tea for breakfast. 3. She tells me she hates it. 4. You might ask her for a cup of it. 5. She loves the cold one.

4. 1. I don't know how. 2. Break a leg 3. My fingers crossed for her. 4. Let's ask her if she likes it. 5. Pardon me

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5. 1. Please follow my instruction. 2. By doing so
3. You don't have to believe me. 4. She will absolutely do it.
5. You have to force her to do it.

6. Situation: Friends are discussing in the classroom.

A : She's upset because of what you said, isn't she?
B : _____.

1. That should work. 2. I can't see the point.
3. That really brought him home. 4. That has nothing to do with it.
5. It doesn't matter.

7. Situation: In the classroom

A: Did you just tell the teacher you are about to be absent from his class?
B: No, _____?
A: You can say that again. Teachers anticipate you to call them when it is necessary to be absent.

1. Would I 2. Could I
3. Should I 4. May I
5. Did I

8. Situation: On phone

Receptionist : Hello, Mandarin Hotel. How may I help you?
Customer : _____.
Receptionist : Yes, please wait a minute.

1. Could you put me through to room 3, please? 2. Could you put the line to room 3, please?
3. Can you call room 3, please? 4. May I speak to room 3, please?
5. Would you mind putting me through to room 3, please?

9. Situation: In the living room

A: The weather is so cold.
B: _____.
A: Oh, that's a good idea.

1. You can't stay outside. 2. You may close the windows.
3. You can turn off the air-conditioner. 4. You should wear a sweater.
5. Let's go outside and chill around.

10. Situation: On the street

A: Sir, can you tell me how to get to the hospital, please?
B: Have you got a car?
A: _____.

1. I would rather take a bus. 2. No, I'm on foot.
3. Yes, I ride it. 4. The car is at my house.
5. I'm exactly sure that I have a car.

Part II: Reading Comprehension (Items 11 – 25)

Scientific Non-Fiction

For 150 years, scientists have tried to determine the solar constant, the amount of solar energy that reaches the Earth. Yet, even in the most cloud-free regions of the planet, the solar constant cannot be measured precisely. Gas molecules and dust particles in the atmosphere absorb and **scatter** sunlight and prevent some wavelengths of the light from ever reaching the ground.

With the advent of satellites, however, scientists have finally been able to measure the Sun's output without being impeded by the Earth's atmosphere. Solar Max, a satellite from the National Aeronautics and Space Administration (NASA), has been measuring the Sun's output since February 1980. Although a malfunction in the satellite's control system limited its observation for a few years, the satellite was repaired in orbit by astronauts from the space shuttle in 1984. Max's observations indicate that the solar constant is not really constant after all.

The satellite's instruments have detected frequent, small variations in the Sun's energy output, generally amounting to no more than 0.05 percent of the Sun's mean energy output and lasting from a few days to a few weeks. Scientists believe these fluctuations coincide with the appearance and disappearance of large groups of sunspots on the Sun's disk. Sunspots are relatively dark regions on the Sun's surface that have strong magnetic fields and a temperature about 2,000 degrees Fahrenheit cooler than the rest of the Sun's surface. Particularly large fluctuations in the solar constant have coincided with sighting of large sunspot groups. In 1980, for example, Solar Max's instruments registered a 0.3 percent drop in the solar energy reaching the Earth. At that time a sunspot group covered about 0.6 percent of the solar disk, an area 20 times larger than the Earth's surface.

Long-term variations in the solar constant are more difficult to determine. Although Solar Max's data have indicated a slow and steady decline in the Sun's output, some scientists have thought that the satellite's aging detectors might have become less sensitive over the years, thus falsely indicating a drop in the solar constant. This possibility was dismissed, however, by comparing Solar Max's observations with data from a similar instrument operating on NASA's Nimbus 7 weather satellite since 1978.

(cited from Free Mind Education)

11. What does this passage mainly discuss?

1. The launching of a weather satellite
2. The components of the Earth's atmosphere
3. The interaction of sunlight and air pollution
4. The solar constant on the Earth's surface
5. The measurement of variations in the solar constant

12. Why does the author mention "gas" and "dust" in line 3?

1. They are parts of the sunlight measurement.
2. They magnify the solar constant.
3. They are found in varying concentrations.
4. Scientific equipment in ruined by gas and dust.
5. They interfere with accurate measurement of the solar constant.

13. Why is it not possible to measure the solar constant accurately without a satellite?

1. The Earth is too far from the Sun.
2. The solar constant is in the dark regions
3. There is not enough sunlight during the day.
4. The Earth's atmosphere interferes with the sunlight.
5. Some areas on Earth receive more solar energy than others.

14. The word "scatter" in line 4 is closest in meaning to _____.

1. emit
2. capture
3. scratch
4. transform
5. disperse

15. According to the passage, scientists believe variations in the solar constant are related to

1. Sunspot group
2. Sunspot activity
3. Unusual weather patterns
4. Increased levels of dust
5. Fluctuations in the Earth's temperature

16. The attempt to describe the solar constant can be best described as_____.

1. an ongoing research effort
2. an issue that has been resolved
3. a question that can never be answered
4. Scientist breakthrough
5. historically interesting, but irrelevant to contemporary concerns.

Complexity

There were two widely **divergent** influences on the early development of statistical methods. Statistics had a mother who was dedicated to keeping orderly records of governmental units (State and statistics come from the same Latin root, status) and a gentlemanly gambling father who relied in mathematics to increase his skill at playing the odds in games of chance. The influence of the mother in the offspring, statistics, is represented by counting, measuring, describing, tabulating, ordering, and the taking of censuses – all of which led to modern descriptive statistics. From the influence of the father came modern inferential statistics which is based squarely in theories of probability.

Descriptive statistics involves tabulating, depicting, and describing collections of data. These data may be quantitative, such as measures of height, intelligence, or grade level – variables that are characterized by an underlying continuum – or the data may represent qualitative variables, such as sex, college major, or personality type. Large masses of data must generally undergo a process of summarization or reduction before they are comprehensible. Descriptive statistics is a tool for describing or summarizing or reducing to comprehensible from the properties of as otherwise unwieldy mass of data.

Inferential statistics is a formalized body of methods for solving another class of problems that present great difficulties for the unaided human mind. This general class of problems characteristically involves attempts to make predictions using a sample of observations. For example, a school system who come to school without breakfast, have been vaccinated for flu, or whatever. Having a little knowledge of statistics, the superintendent would know that it is unnecessary and inefficient to question each child; the proportion for the entire district could be estimated fairly accurately from a sample of as 100 children. Thus, the purpose of inferential statistics is to predict or estimate characteristics of a population from a knowledge of the characteristics of only a sample of the population.

(cited from Free Mind Education)

17. What is the passage mainly concerned?

1. The drawbacks of descriptive and inferential statistics
2. Applications of inferential statistics
3. The development and use of statistics
4. How to use descriptive statistics
5. The implementation of statistics

18. The word "divergent" in line 1 is closest meaning to _____.

1. classify
2. different
3. distributed
4. recorded
5. prominent

19. According to the first paragraph, counting and census-taking are associated with _____.

1. Inferential statistics
2. Descriptive statistics
3. Unknown variables
4. Quantitative changes
5. Qualitative changes

20. Which of the following is NOT given as an example of a qualitative variable?

1. Gender
2. Height
3. Demographic
4. College major
5. Type of personality

21. Which of the following statements about descriptive statistics is best supported by the passage?

1. It tabulates the data.
2. It reduces large amounts of data to a more comprehensible form.
3. It is based on probability.
4. It can be used by people with little knowledge of mathematics.
5. It measures only qualitative differences.

Black Holes

Nothing in the history of modern astronomy has excited as much speculation as the object, or event, known as a black hole. Black holes have provided endless imaginative fodder for science fiction writers and endless theoretical fodder for astrophysicists. They are one of more exotic manifestations of the theory of general relativity, and their fascination lies in the way their tremendous gravity affects nearby space and time.

A black hole is very simple in structure: it has a surface – the event horizon – and a center – the singularity. Everything else is gravity. The standard model for the formation of a black hole involves the collapse of a large star. The imaginary spherical surface surrounding the collapsed star is the event horizon – an artificial boundary is space that marks a point of no return. Outside the event horizon, gravity is strong but finite, and it is possible for objects to break free of its pull. However, once within the event horizon, an object would need to travel faster than light to escape.

For extremely massive stars, the exclusion principle – the resistance between the molecular particles within the star as they are compressed – will not be strong enough to offset the gravity generated by the star's own mass. The star's increasing density will overwhelm the exclusion principle. What follows is runaway gravitational collapse. With no internal force to stop it, the star will simply continue to collapse in on itself. Once a collapsing star has contracted through its event horizon, nothing can stop it from collapsing further until its entire mass is crushed down to a single point – a point of infinite density and zero volume – the singularity.

The star now disappears from the perceivable universe, like **a cartoon character** that jumps into a hole and pulls the hole in after him. What this process leaves behind is a different kind of hole – a profound disturbance in space – time, a region where gravity is so intense that nothing can escape from it. Any object falling within the boundary of a black hole has no choice but to move inward toward the singularity and disappear from our universe forever. Moreover, a black hole can never be plugged up or filled in with matter; the more matter that is poured into a black hole, the bigger it gets.

What would happen to objects, such as astronauts, as they vanished into a black hole? Physicists have been amusing themselves with this question for years, and most believe that the intense gravitational forces would rip apart the astronauts long before they were crushed at the singularity. Theoretically, any astronauts who managed to survive the passage would encounter some very strange things. For instance, they would experience acute time distortion, which would enable them to know, in a few brief seconds, the entire future of the universe.

Inside a black hole, space and time are so warped that the distance from the event horizon to the singularity is not a distance in space in the normal sense that we can measure in kilometres. Instead, it becomes a distance in time. The time it takes to reach the singularity from the event horizon – as measured by someone falling in – is proportional to the mass of the black hole.

22. What happens to an object that falls within the event horizon of a black hole?

1. The object changes shape until it is spherical.
2. The object is pushed from the hole at the speed of light.
3. The object explodes into particles that drift into space.
4. The object disappears into the singularity.
5. The object cannot escape the black hole's gravity.

23. Why dies the author mention **a cartoon character** in paragraph 4?

1. To point out a humorous phenomenon in astrophysics
2. To announce the creation of a cartoon about black holes
3. To describe the creation of black holes.
4. To warn of the danger of being sucked into a black hole.
5. To illustrate the complete disappearance of a collapsing star

24. Astronauts who fell into a black hole would probably experience all of the following EXCEPT

1. knowledge of singularity
2. distortion of space and time
3. traveling faster than light
4. knowledge of the universe
5. strong gravitational forces

25. What can be inferred from paragraph 6 about the distance between the event horizon and the singularity?

1. The distance increases and decreases continuously
2. The distance is related to the size of the black hole.
3. The distance is more than several trillion kilometres.
4. The distance cannot be traveled in less than a year.
5. The distance needs to travel faster than light.

Part III: Structure and Writing (Items 26 – 40)

26. Thailand has an aging society with _____.

1. senior citizens of the population making up about 15%
2. about 15% of the population makes up senior citizens
3. making up of about 15% senior citizen population
4. senior citizens making up about 15% of the population
5. the population makes up of about 15% senior citizens

27. Even with vast research, there is still a great deal that is _____ known about the hidden secret of the human mind.

1. none
2. neither
3. not
4. no
5. nor

28. A stanza is a short sentence composition written especially _____ a particular technique.

1. enable students practicing
2. enables students practicing
3. enable students to practice
4. to enable students to practice
5. enabling students to practice

29. _____ advance and retreat in their eternal rhythms, but the surface of the sea itself is never at rest.

1. When the tides not only do
2. Not only when the tides do
3. As the tides not only do
4. Not only do the tides
5. Do the tides not only

30. In Thailand, a primary election is a method _____ voters select the representatives for public office.

1. that
2. is that
3. by which
4. by those
5. in which

31. Most reptiles hatch from eggs laid in water or moist ground, and begin life _____ water-dwelling larvae.

1. are
2. such
3. as
4. to be
5. such as

32. Nearly all trees contain a mix of polymers that can burn like petroleum _____ properly extracted.

1. after
2. is
3. if
4. when it
5. which

33. Simple camera recorder cannot _____ sharp, undistorted footages over a wide field.

1. to form
2. forming
3. from
4. are formed
5. be formed

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34. _____ of humans is known as anthropology.

1. The scientific study
2. Studying scientific
3. That is scientific studying
4. To study scientifically
5. The study of scientific

35. In bacteria and in other organisms, _____ is the nucleic acid DNA that provides the genetic information.

1. which
2. both
3. it
4. and
5. where

Directions: Choose the answer which is grammatically <u>incorrect</u>.

36. **(1)** **<u>To rid</u>** the school parks of plastic, foam, and **(2)** **<u>other non-degradable packagings</u>**, students **(3)** **<u>will be asked</u>** to use **(4)** **<u>environmental-friendly</u>** products such as cassava **(5)** **<u>lunch boxes</u>**.

37. The mayor **(1)** **<u>surprisingly</u>** addressed the **(2)** **<u>massively</u>** audience in the **(3)** **<u>city</u>** auditorium without **(4)** **<u>looking at</u>** her **(5)** **<u>numerous</u>** notes.

38. Even **(1)** **<u>granted extenuating circumstances</u>**, the gardener **(2)** **<u>still might have</u>** managed **(3)** **<u>to bring</u>** in the **(4)** **<u>fragile</u>** tree palm and rose bush **(5)** **<u>before</u>** the storm.

39. **(1)** **<u>Having appointed</u>** to the Defense Ministry in 2005, Bangor Aldoras **(2)** **<u>has served</u>** as Vice President for Interior Affairs between 2010 – 2020, **(3)** **<u>during which time</u>** he **(4)** **<u>helped</u>** design the **(5)** **<u>multi-billion</u>** infrastructure project.

40. A **(1)** **<u>centuries old</u>** mountaintop villa in central Indonesia **(2)** **<u>caught fire</u>**. It was unclear **(3)** **<u>how much</u>** of its **(4)** **<u>structures</u>** were damaged by the fire. **(5)** **<u>Nevertheless</u>**, no one was injured.

Part IV: Vocabulary (Items 41 – 50)

<u>Complete the sentence with the appropriate vocabulary</u>

41. The motorbike repair shop considered the scratches on the left side of the motorbike to be caused by a major_____.

1. conquest
2. mishap
3. jovial
4. unavoidable
5. uphold

42. Although mammoths have been_____for thousands of years, they are still_____among present youth.

1. sporadic - special
2. extinct - popular
3. unknown - critical
4. medium - encouraged
5. mighty - relaxed

43. The_____force of the Tsunami will abruptly_____the state with water.

1. neutral - destroy
2. powerful - inundate
3. mighty - overwhelm
4. catastrophic - battle
5. fierce - blow

44. Since our workplace was so busy and_____, we longed most of all for_____.

1. hardworking - employee
2. neat - olfactory
3. proclamation - liability
4. fulcrum - candid
5. noisy - solitude

45. Many laboratories have been_____how air pollution being contaminated could_____the issue in the future.

1. alleviating - deteriorate
2. issuing - enlarge
3. evaluating - aggravate
4. purifying - damage
5. defying - exceed

Choose the choice which provides the same meaning as the underlined word in the given sentence.

46. The bank of the Nile river is fertile of minerals and creatures.

1. English athletes annually swim across the opposite bank of the Channel.
2. The Moon appeared behind a bank of cloud.
3. A blood bank is considered as a center of blood donation.
4. The River Bank is situated in the downtown of London.
5. The bank of switches remains in a good condition.

47. The Icon Siam marks another global-scale event in Thailand.

1. The employees need eight hours to mark all these mechanical forms.
2. The teacher marks my book before submitting to the principal.
3. You can mark my words, he is cheating on you.
4. It is elegance that marks Vatanika dressing style.
5. We never get a good mark in her subject.

48. Thousands of school flyers have been delivered to all students by the teachers.

1. The judge eventually delivered a verdict of being guilty.
2. The mayor of the town delivered the speech in front of the citizen.
3. My aunt who is a teacher at the school has been delivered of a healthy baby girl.
4. The food has been delivered to the front gate of my house.
5. The president of the school has delivered the keynote address.

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49. At the end of each month, my father pays expense bill by online banking since it is very convenient.

1. I pay 2,000 Baht a week for the internet usage.
2. Thais pay respect to their King Rama Nine in the ceremony.
3. I always pay attention to the way of managing expense bill.
4. She should pay for her mistakes in this project.
5. The government pays back the corruption according to the bill.

50. We were charged with criminal offences at the police station after we reluctantly follow the traffic police instruction.

1. The Revenue Department charged fee to change the company's address.
2. The policeman were charged with taking care of the bank.
3. Two burglars are charged with assault two days ago.
4. The river in India was charged with contaminated chemical from the ship.
5. The judges were charged with choosing the best model in the beauty contest.

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